

Bottom-up Construction of Universally Baire Sets

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Axiom

There exists a proper class of Woodin limit of Woodin cardinals.



Theorem (Woodin)

1. Every set of reals in CM^+ is universally Baire.
2. For all $X \in \text{CM}^+$, μ_X is a CM^+ -ultrafilter.
3. $(\text{CM}^+)^\sharp$ exists and is coded by a universally Baire set.



Theorem (Sargsyan)

1. There exists the $\subseteq$ -minimum inner model containing all reals and satisfying $\text{AD}_{\mathbb{R}} + \text{"}\Theta \text{ is regular"}$.
2. All sets of reals in this model are universally Baire.



Theorem (K., Sargsyan)

The minimum model of $\text{AD}_{\mathbb{R}} + “\Theta \text{ is regular}”$ is contained in the Chang-plus model.



Theorem (K., Sargsyan)

$L((\text{CM}^+)^\sharp)$ contains an iteration strategy for a hod mouse with a measurable limit of Woodin cardinals.



- ▶ We are interested in the preordering $(\mathcal{P}(\mathbb{R}), \leq_w)$.
- ▶ More specifically, we want to understand its subset UB, which consists of all *universally Baire* sets of reals.

Theorem (Martin, Steel, Wadge, Woodin)

UB is well-founded, linear, and downwards closed in $(\mathcal{P}(\mathbb{R}), \leq_w)$.



- Increasing initial segments of $(\text{UB}, \leq_W)$: $\Sigma_0^1, \Sigma_1^1, \Sigma_2^1, \dots$,

$$\Sigma_\omega^1 = L_1(\mathbb{R}) \cap \mathcal{P}(\mathbb{R}),$$

$$L_2(\mathbb{R}) \cap \mathcal{P}(\mathbb{R}), \dots, L_\alpha(\mathbb{R}) \cap \mathcal{P}(\mathbb{R}), \dots, L(\mathbb{R}) \cap \mathcal{P}(\mathbb{R})$$

- How do we proceed? (How do we get $\mathbb{R}^\#$?)

Definition

Let $\Delta_0 := L(\mathbb{R}) \cap \mathcal{P}(\mathbb{R})$.



Definition

For $\Gamma \triangleleft (\text{UB}, \leq_W)$, let $\mathcal{O}_1(\Gamma) := L(\Gamma^\omega) \cap \mathcal{P}(\mathbb{R})$. □

Fact

Let $\Gamma \triangleleft \text{UB}$. Then the following holds.

- ▶ $\Gamma \trianglelefteq \mathcal{O}_1(\Gamma) \triangleleft \text{UB}$
- ▶ $\Gamma \triangleleft \mathcal{O}_1(\Gamma)$ if and only if $\text{cof}(\Gamma, \leq_W) = \omega$. □

Theorem

It holds that $\text{cof}(\Delta_0) = \omega$. In particular, $\Delta_0 \triangleleft \mathcal{O}_1(\Delta_0) \triangleleft \text{UB}$. □

- ▶ So, $\mathbb{R}^\# \in \mathcal{O}_1(\Delta_0)$.

Definition

For $\mathcal{O} : \{\Gamma \triangleleft \text{UB}\} \longrightarrow \{\Gamma \triangleleft \text{UB}\}$, define $\mathcal{O}^\alpha(\Gamma)$ as follows:

1. $\mathcal{O}^0(\Gamma) := \Gamma$
2. $\mathcal{O}^{\alpha+1}(\Gamma) := \mathcal{O}(\mathcal{O}^\alpha(\Gamma))$
3. for γ limit, $\mathcal{O}^{<\gamma}(\Gamma) := \bigcup_{\alpha < \gamma} \mathcal{O}^\alpha(\Gamma)$ and

$$\mathcal{O}^\gamma(\Gamma) := L(\mathcal{O}^{<\gamma}) \cap \mathcal{P}(\mathbb{R}).$$

Fact

There is α such that $\mathcal{O}_1^\alpha(\Delta_0) = \mathcal{O}_1^{\alpha+1}(\Delta_0) \triangleleft \text{UB}$.



- ▶ Let Δ_1 be the shortest $\Gamma \triangleleft \text{UB}$ satisfying $\mathcal{O}_1(\Gamma) = \Gamma$.
- ▶ There exists α is such that $\mathcal{O}_1^\alpha(\Delta_0) = \Delta_1$.
- ▶ How do we continue beyond Δ_1 ?

Definition

For X a set, let μ_X be the club filter on $[X]^\omega$. □

Definition

For $\Gamma \triangleleft \text{UB}$, let $\mathcal{O}_2(\Gamma) := L(\Gamma^\omega)[\mu_\Gamma] \cap \mathcal{P}(\mathbb{R})$. □

Theorem

For $\Gamma \triangleleft \text{UB}$, $\mathcal{O}_2(\Gamma) \triangleleft \text{UB}$. □

Theorem (Woodin)

$\Delta_1 \triangleleft \mathcal{O}_2(\Delta_1) \triangleleft \text{UB}$ □

Fact

There exists α such that $\mathcal{O}_2^\alpha(\Delta_1) = \mathcal{O}_2^{\alpha+1}(\Delta_1) \triangleleft \text{UB}$. □

- ▶ Let Δ_2 be the shortest $\Gamma \triangleleft \text{UB}$ such that $\mathcal{O}_2(\Gamma) = \Gamma$.
- ▶ For some α , $\Delta_2 = \mathcal{O}_2^\alpha(\Delta_1)$.
- ▶ How do we continue?

Theorem (Ikegami, Trang)

Let $\Gamma \triangleleft \text{UB}$ be L -closed. If $\mathcal{O}_2(\Gamma) = \Gamma$, then $\text{cof}(\Gamma) > \omega$ and Γ has the scale property. □

- ▶ HOD Analysis à la Steel applies to Δ_2 !

Definition

For $\Gamma \triangleleft \text{UB}$ and $\kappa \leq \text{Ord}$, let

$$\mathcal{O}_3^*(\Gamma, \kappa) := L(\Gamma^\omega, \kappa^\omega)[\mu_\Gamma, \mu_\kappa] \cap \mathcal{P}(\mathbb{R}).$$



Definition

For $\Gamma \triangleleft \text{UB}$, let $\mathcal{O}_3(\Gamma)$ be defined as $\mathcal{O}_3^*(\Gamma, \kappa)$ for the smallest κ satisfying that $\Gamma \triangleleft \mathcal{O}_3^*(\Gamma, \kappa)$ or $\kappa = \text{Ord}$.



Theorem (K., Sargsyan)

$$\Delta_2 \triangleleft \mathcal{O}_3(\Delta_2) \triangleleft \text{UB}$$



Theorem (K., Sargsyan)

Suppose

1. $\Gamma \triangleleft \text{UB}$ is L -closed,
2. $\text{cof}(\Gamma) > \omega$ and Γ has the scale property,
3. no set in Γ codes an iteration strategy for a hod mouse with a measurable limit of Woodin cardinals.

Then $\Gamma \triangleleft \mathcal{O}_3(\Gamma) \triangleleft \text{UB}$.



Corollary

For some α , $L(\mathcal{O}_3^\alpha(\Delta_2))$ is the minimum model of $\text{AD}_{\mathbb{R}} + \text{“}\Theta \text{ is regular”}$.



- Chang-plus model is $\text{CM}^+ := L(\text{Ord}^\omega)[(\mu_{\alpha^\omega} : \alpha \in \text{Ord})]$.

Theorem (Woodin)

1. Every set of reals in CM^+ is universally Baire.
2. For all $X \in \text{CM}^+$, μ_X is a CM^+ -ultrafilter.
3. $(\text{CM}^+)^\sharp$ exists and is coded by a universally Baire set. □

Corollary (K., Sargsyan)

The minimum model of $\text{AD}_{\mathbb{R}} + “\Theta \text{ is regular}”$ is contained in the Chang-plus model. □

Corollary (K., Sargsyan)

$L((\text{CM}^+)^\sharp)$ contains an iteration strategy for a hod mouse with a measurable limit of Woodin cardinals. □

Fact

For some α , $\mathcal{O}_3^\alpha(\Delta_2) = \mathcal{O}_3^{\alpha+1}(\Delta_2) \triangleleft \text{UB}$. □

- ▶ Let Δ_3 be the shortest $\Gamma \triangleleft \text{UB}$ such that $\mathcal{O}_3(\Gamma) = \Gamma$.
- ▶ How do we continue? What is “ $\mathcal{O}_4$ ”?
- ▶ Future work will show!